Benjamin Andrew

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github.com/benmandrew

Education

University of Manchester, PhD in Computer Science — Sep 2024 – Present

- Project titled *Developing Reliable Software for Autonomous Robots*. Supervised by Louise Dennis, Marie Farrell, and Michael Fisher.

University of Cambridge, BA (Hons) in Computer Science, II.1 — Oct 2019 – Jun 2022

- Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by Dr. Jackson Woodruff.

Teaching

Graduate Teaching Assistant, University of Manchester — Sep 2024 – Present

- Logic and Modelling (2nd-year course)

Supervisor (TA equiv.), University of Cambridge — Oct 2022 – Jun 2024

- Logic and Proof (2nd-year course)
- Hoare Logic and Model Checking (3rd-year course)
- Semantics of Programming Languages (2nd-year course)

Professional Experience

Software Engineer, Tarides – Paris, France — Sep 2022 – Jun 2024

- Developed open-source CI (Continuous Integration) software for the OCaml language's ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
- Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.

Embedded Systems Intern, Tarides – Paris, France — Jul 2021 – Sep 2021

- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.

Computer Graphics Research Intern, University of Cambridge — Jun 2020 – Sep 2020

- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Volunteering

English Teacher, Connections English Institute – Ibagué, Colombia — Jul 2024 – Aug 2024

- Taught adult beginners in morning and night classes.

Related Personal Projects

Cavalry — Imperative mini-language in which programs are formally verified according to logical user specifications using a Hoare logic-based prover. Supports procedures, allowing compositional and reusable proofs (*link*).

Probabilistic Model Checker — Verifier for probabilistic CTL statements about Markov models (*link*).